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SECTION 02

GROUP 3

ANALYSIS AND DESIGN OF A BUSINESS INTELLIGENCE DASHBOARD
FOR MANUFACTURING PERFORMANCE

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1.0 Main Problem

1.1 Introduction

PT. MPM is a packaging manufacturing company that produce both mass-produced and customized packaging products. The company collects huge amount of operational data from various departments such as Sales, Manufacturing, Inventory and Warehouse. These data are crucial for measuring organizational performance and for business decision making.

However, there is no integrated system for measuring and monitoring the performance of the core business processes at PT. MPM. Information is stored in silos within departments, making it hard for management to get a clear and accurate picture of the organization's performance. This makes decision-making processes inefficient and time consuming.

1.2 Problems identified

1. Lack of Performance Monitoring System

The company does not have a single platform for monitoring and evaluating the performance of the business. If there is no monitoring system, then management will have problems in evaluating the effectiveness of sales and production activities.

2. Data Fragmentation Across Departments

Each departments keeps its own operational data, like sales reports, production report, inventory records and machine performance data. This disjointed data structure makes it difficult to retrieve and analyze the information.

3. Difficulty in Monitoring Key Performance Indicators (KPIs)

Management cannot effectively monitor key business metrics such as revenue, forecast sales, quantity sold, actual production and others. It lacks one central dashboard reduces visibility of organizational performance.

4. Inefficient Decision-Making

Management needs to gather and process data from several sources to make business decisions. This takes a lot of time and can postpone strategic decisions.

5. Reduced Competitiveness

Failure to detect operational problems and trends in performance promptly means that the company is less responsive to market changes and less competitive.

2.0 Main Solution

The main solution proposed to address the performance monitoring challenges at PT.MPM is the development of a Business Intelligence (BI) Dashboard System. This system integrates data from multiple departments into a centralized platform, enabling management to monitor organizational performance effectively and make data-driven decisions in a timely manner.

2.1 Business Intelligence Dashboard

The BI Dashboard serves as the primary solution to the lack of a performance monitoring system at PT.PMP. It provides top management with a single, unified interface to view the key business metrics across both sales and production operations. The Sales Performance Dashboard and the Production Performance Dashboard are two specialized dashboards that allows management to view all essential data without having to manually compile reports from several departments.

The Sales Performance Dashboard allows management to assess the sales performance against KPIs such as revenue, forecast sales versus actual sales, product performance and customer performance. The dashboard is equipped with interactive filters for date range, quarter, customer ID and product category, allowing the flexibility to focus on specific time periods or customer segments. Besides, key visualizations also include pie charts that break down product and customer performance, a heatmap table that use color-codes performance rankings for forecast versus actual sales comparison and a combo chart that shows contrast the datasets.

The Production Performance Dashboard enables management to track the Overall Equipment Effectiveness (OEE) across each production line and evaluate productivity of the production department. Management can filter data by date range, product division and machine to narrow down their analysis to specific production sections. The dashboard features gauge charts that visually display the three OEE components which are performance ratio, availability ratio and quality ratio alongside a bar chart that illustrates the difference between planned and actual production output. Furthermore, two scorecards summarize the total planned and actual

production for selected period, while the heatmap pivot table allows management to evaluate OEE performance for each individual machine.

2.2 ETL Process

The ETL (Extract, Transform, Load) process assembles fragmented departmental data into a unified and structured format. Data is collected from diverse sources across the organization, where the primary phase involves converting all gathered data into Google Sheets format to ensure consistency. The ETL process then initiates data cleaning, where superfluous and irrelevant data is eliminated and only retained the essential data. During the transformation phase, necessary adjustments are made such as converting string-formatted date files into proper date formats. For the production performance table, data is sourced from both the production report and the maintenance report before ETL process commences. Once all transformations are completed, the cleaned and structured data is loaded into the data warehouse, producing a warehouse database that fully satisfies the system requirements.

2.3 Data Warehouse

The data warehouse serves as the central repository that stores all integrated organizational data from both the Sales and Manufacturing departments in Google Sheets format, which connects directly to Google Data Studio for dashboard visualization. By consolidating data from multiple departments into a single database, it eliminates data fragmentation and provides a single source of truth for the entire organization, storing both sales-related data such as revenue, forecast sales and quantity sold, as well as production-related data such as OEE, planned production, actual production, machine performance, availability and quality.

2.4 Google Data Studio as the Development Tool

Google Data Studio was selected as the platform for developing and hosting the BI Dashboard for PT.MPM. It is a web-based, user-friendly tool that enables the creation of interactive and visually appealing data visualizations without requiring advanced programming knowledge. Since it is cloud-based, authorized users can access the dashboard from any device with internet connection and its wide range of built-in visualization types. For instance, scorecards, gauge charts, heatmaps, pie charts, bar charts and combo charts that allows complex organizational data to be presented in a clear and understandable format for management.

2.5 KPI Monitoring

Management at PT.MPM can now continuously monitor sales and production KPIs through a single, centralised platform as a result of the implementation of BI dashboard. Sales KPIs such as tool revenue, forecast accuracy, quantity sold and customer performance, as well as production KPIs such as OEE, machine availability, performance ratio and quality ratio are all visible at glance. This real-time visibility enables management to quickly identify underperforming areas and take corrective action promptly, ultimately improving operational efficiency and strengthening PT.MPM's competitiveness in the market.

3.0 User Stories

3.1 Top Management

User Story 1:

As a Top Manager, I want to identify underperforming products so that I can plan improvement strategies.

Description:

The Top Manager uses the dashboard to analyze the product performance using sales and production information. Management can make better decisions about product improvement, marketing strategies, or resources when they identify products that are not meeting their customers' expectations.

3.2 Sales Department

User Story 2:

As a Sales Officer, I want to monitor customer purchasing trends so that I can improve sales planning.

Description:

Sales Officer should have access to customer purchasing patterns and sales trends in the dashboard. This information provides the sales team with insight into customer preferences, future needs, and better sales strategies.

3.3 Production Department

User Story 3:

As a Production Manager, I want to compare planned production with actual production so that I can evaluate production efficiency.

Description:

The Production Manager checks the dashboard for comparison of production targets and actual production. This enables the manager to know what he is lacking and keep track of how things are going and make necessary adjustments to make things better.

3.4 Data Analyst

User Story 4:

As a Data Analyst, I want to collect and integrate data from different departments so that the dashboard provides accurate information.

Description:

The job of the Data Analyst is to extract, convert and load data from the sales and production departments into the data warehouse. Consistent and accurate data guarantees that the dashboard gives consistent and accurate information for decision making.

4.0 Use Case Diagram

Figure 4.1 illustrates the Use Case Diagram of the proposed Business Intelligence Dashboard System developed for PT. MPM. The diagram presents the interactions between system users and the functionalities provided by the dashboard. The system is designed to support performance monitoring, data integration and business decision-making through centralized sales and production dashboards.



Figure 4.1: Use Case Diagram of Business Intelligence Dashboard System

4.1 Description of Use Case Diagram

Top Management

Top Management is the primary user of the dashboard system. This actor utilizes the dashboard to monitor organizational performance and support strategic decision-making. Top Management can access the Sales Dashboard and Production Dashboard to evaluate business performance. In addition, the actor can monitor Key Performance Indicators (KPIs), filter

dashboard information based on specific criteria and generate reports for performance evaluation.

The View Sales Dashboard and View Production Dashboard use cases include the Filter Dashboard Data use case. This relationship indicates that management can filter information by date, customer, product category or production period to obtain more specific insights.

Data Analyst

The Data Analyst is responsible for managing and preparing organizational data before it is presented through the dashboard. The actor performs activities such as uploading data, cleaning data, transforming data and updating the data warehouse.

The use cases reflect the Extract, Transform, Load process described in the case study, where data is extracted from multiple departments, cleaned and transformed into a consistent format and then loaded into the data warehouse.

Sales Department

The Sales Department contributes sales-related information to the system. The actor submits sales data, including revenue, forecast sales, quantity sold, customer information and product information. The department can also access the Sales Dashboard to monitor sales performance and evaluate sales trends.

Manufacturing Department

The Manufacturing Department provides production-related information to the system. The actor submits production data such as planned production, actual production, machine performance, availability, quality and Overall Equipment Effectiveness (OEE). The department can also access the Production Dashboard to monitor production performance and operational efficiency.

5.0 ERD Diagram (UML Notation)

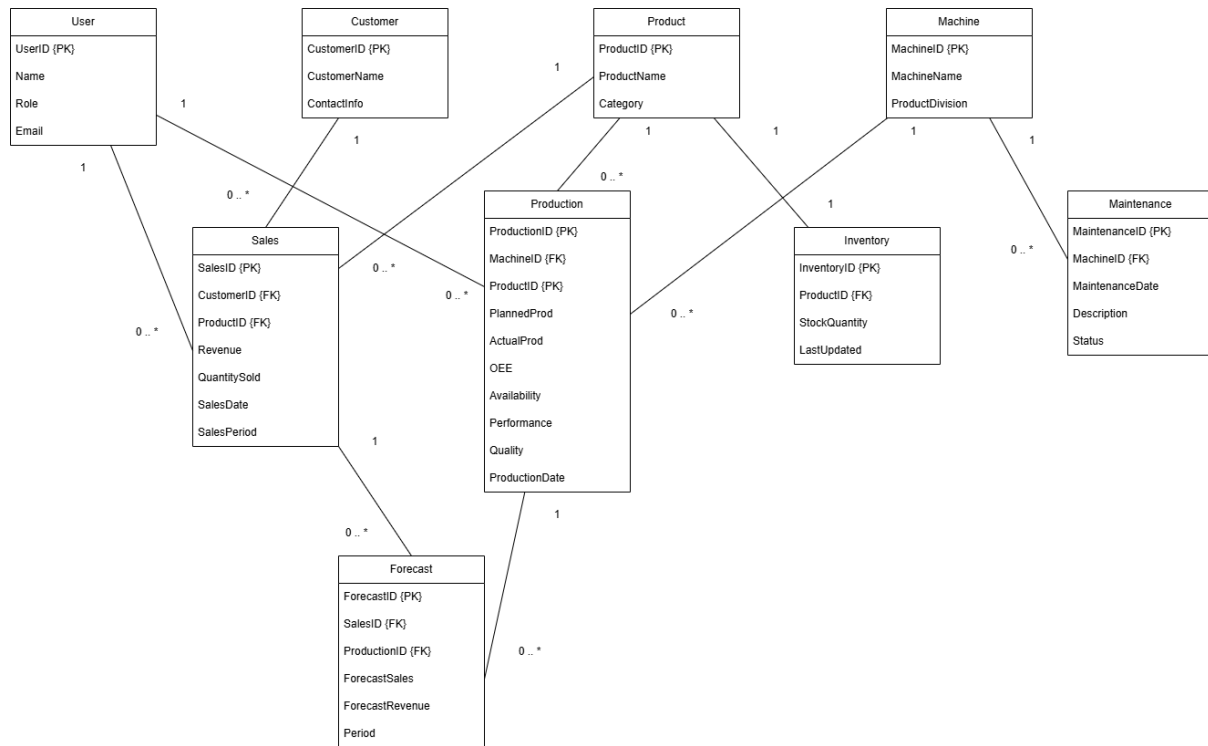


Figure 5.1: ERD Diagram

USER: This entity represents system users with different roles such as Top Management, Data Analyst, Sales Officer and Production Manager. They interact with the system by viewing sales and production dashboards, monitoring KPIs and generating reports based on their access level.

CUSTOMER: This entity represents clients who purchase products from PT. MPM. Customer information is linked to sales transactions to enable analysis of customer purchasing trends and performance.

PRODUCT: This entity represents the products manufactured and sold by PT. MPM, including product name and category. Each product is identified by a unique ProductID and connects to the Sales, Production and Inventory entities, making it a central entity that links the sales and production sides of the business.

SALES: This entity records all sales transactions including revenue, quantity sold, sales date and period. It links to both the Customer and Product entities through foreign keys, allowing management to analyse sales performance by product category, customer and time period.

FORECAST: This entity stores forecast targets for both sales and production. It is linked to Sales and Production to enable comparison between projected and actual values. This supports management in evaluating forecast accuracy and identifying gaps in sales and production performance.

MACHINE: This entity represents the production machines used in PT. MPM's production lines. Each machine is identified by a unique MachineID and is associated with a machine name and product division. Production Division refers to the category of production line that a machine is assigned to manufacture. It is linked to both the Production and Maintenance entities, making it a key entity for monitoring machine-level performance and service history.

PRODUCTION: This entity records production performance data per machine including planned vs actual production, OEE, availability, performance and quality ratios. By linking to both the Machine and Product entities, this entity allows management to evaluate production efficiency at both the machine level and product division level.

INVENTORY: This entity tracks product stock levels in the warehouse. It stores the stock quantity and last updated date for each product and links directly to the Product entity through a foreign key. This allows the organization to monitor inventory availability and ensure production and sales operations are supported by sufficient stock levels.

MAINTENANCE: This entity records machine maintenance history including maintenance date, description and status. It stores information such as maintenance date, description of the maintenance carried out and the current status. By linking to the Machine entity, this entity allows the manufacturing department to track each machine's service records and monitor the impact of maintenance on production performance.

6.0 Data/System Architecture

The Data/System Architecture illustrates the flow of data within the proposed Business Intelligence Dashboard System developed for PT. MPM. The architecture integrates data from multiple departments and transforms raw operational data into meaningful information that supports business decision-making.

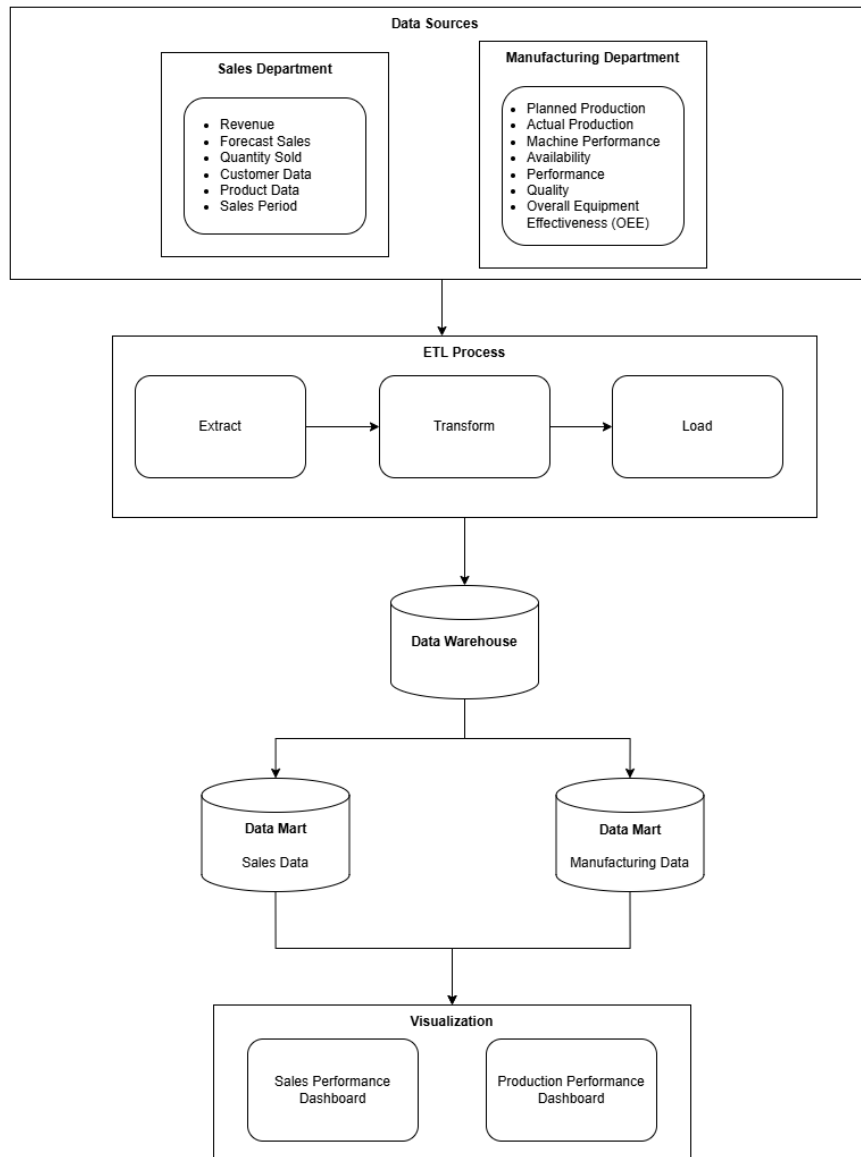


Figure 6.1: System Architecture of the Business Intelligence Dashboard System

6.1 Overview of the Architecture

The proposed architecture consists of several components, namely Sales Department, Manufacturing Department, Data Analyst, ETL Process, Data Warehouse, Business Intelligence Dashboard System, and Top Management. These components work together to collect, process, store, and visualize organizational data.

The architecture is designed to ensure that business information is centralized and easily accessible for performance monitoring and strategic decision-making.

6.2 Data Sources

The first layer of the architecture consists of operational data generated by the Sales Department and Manufacturing Department.

	Sales Department	Manufacturing Department
Information	○ Product information ○ Customer information ○ Revenue ○ Forecast sales ○ Quantity sold ○ Sales period	○ Planned production ○ Actual production ○ Machine performance ○ Availability ○ Quality ○ Performance ratio ○ Overall Equipment Effectiveness (OEE)
Data Usage	to evaluate sales performance and monitor business growth	to monitor production efficiency and operational performance

6.3 ETL Process

After data collection, the information passes through the ETL process.

Extract - Data is extracted from multiple departmental sources.

Transform - The extracted data is cleaned and transformed into a consistent format. Duplicate, incomplete, or unnecessary records are removed to improve data quality.

Load - The processed data is loaded into the data warehouse for storage and future analysis.

The ETL process ensures that only accurate and reliable information is available for reporting and visualization purposes.

6.4 Data Warehouse

The Data Warehouse serves as the central repository for storing integrated organizational data. Information from different departments is consolidated into a single database, eliminating data silos and improving accessibility.

The data warehouse contains both sales and production information that can be used for business analysis and KPI monitoring.

6.5 Data Mart

To support specific business analysis requirements, the Data Warehouse is divided into two specialized Data Marts.

Sales Data Mart

The Sales Data Mart stores sales-related information and supports sales performance analysis, including revenue monitoring, sales forecasting, customer analysis, and product performance evaluation.

Manufacturing Data Mart

The Manufacturing Data Mart stores production-related information and supports manufacturing performance analysis, including production output, machine performance, availability, quality, and Overall Equipment Effectiveness (OEE).

The use of Data Marts improves query performance and allows users to focus on department-specific information.

6.6 BI Dashboard

The Business Intelligence Dashboard System retrieves information from the data warehouse and presents it through interactive visualizations.

The dashboard consists of two main components:

Sales Performance Dashboard

The Sales Performance Dashboard enables management to monitor:

- Revenue
- Forecast sales
- Actual sales
- Product performance
- Customer performance

Production Performance Dashboard

The Production Performance Dashboard enables management to monitor:

- Overall Equipment Effectiveness (OEE)
- Availability
- Performance
- Quality

- Planned production
- Actual production

These dashboards provide real-time visibility into organizational performance and support data-driven decision-making.

7.0 Multidimensional Design

A Galaxy Schema is a multidimensional data model where more than one fact table shares one or more dimension tables. It is widely used in Business Intelligence systems that have to monitor several business processes of the same company. The company keeps a track of both sales and production activities in this case study. Thus, a Galaxy Schema is proposed to merge data of both business areas, while enabling common dimensions like Product and Date.

7.1 Galaxy Schema Diagram

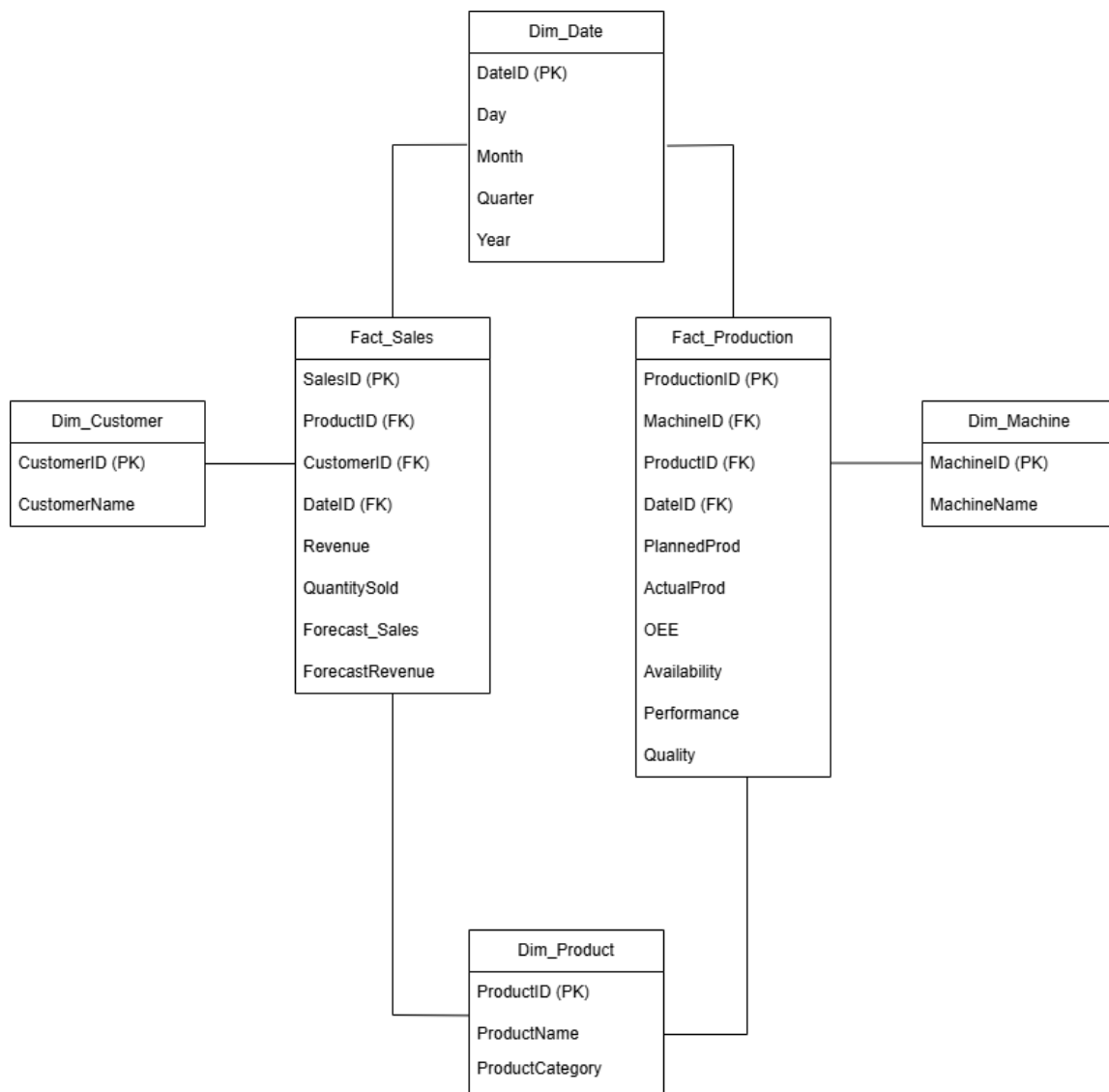


Figure 7.1 Galaxy Schema Diagram

7.2 Explanation of Galaxy Schema Diagram

The proposed Galaxy Schema aims to be able to analyze sales and production performance using one Business Intelligence (BI) system. It is comprised of two fact tables: Fact_Sales and Fact_Production, that contain the key performance data that is important to management for monitoring and decision making.

The Fact_Sales table has the sales information including revenue, quantity sold, forecast sales, and forecast revenue. These measures enable management to check how sales are performing in relation to targets and track the company's growth over time.

The Fact_Production table includes measurements for production, such as planned production, actual production, Overall Equipment Effectiveness (OEE), availability, performance, and quality. These indicators enable management to evaluate the efficiency of production and pinpoint areas in which operation can be improved.

There are multiple dimension tables that are common to the fact tables. The Product dimension contains useful details on the products sold and produced, and the Date dimension is used to analyse over time. Customer is used for sales analysis while Machine is used for production analysis.

The company can now combine sales and production data in one analytical model with the use of a Galaxy Schema. This provides support for dashboard reporting and allows management to have a 360 view of business performance to aid decision making.